

Proposed Solution

Project: Employee Software Installation Request (ServiceNow)

1. Solution Overview

The proposed solution is a **ServiceNow-based Software Installation Request Management System** that automates the complete software request lifecycle. Employees can request licensed software through a standardized **Service Catalog** form, eliminating the need for emails or verbal communication.

Once a request is submitted, ServiceNow automatically routes it to the employee's reporting manager for approval. If approved, a **Catalog Task** is created and assigned to the **Network/Software Support Team** for software installation. If the requested software cannot be installed due to license unavailability, the system automatically generates an **Incident** to ensure the issue is tracked and resolved.

This workflow provides a centralized, transparent, and efficient process for handling software installation requests while improving communication and accountability among employees, managers, and IT teams.

2. Project Objectives

The proposed solution aims to achieve the following objectives:

- Replace manual email-based software requests with a standardized and automated request process.
 - Provide employees with an easy-to-use Service Catalog form for submitting software installation requests.
 - Enable managers to review, approve, or reject requests through the **My Approvals** interface.
 - Automatically assign approved requests to the Network/Software Support Team for fulfillment.
 - Notify employees about important updates throughout the request lifecycle.
 - Automatically detect license shortages and create incidents for timely investigation and resolution.
 - Improve transparency, accountability, and overall efficiency of the software request process.
-

3. Solution Components

The solution is built using the following ServiceNow components.

Component	Role in the Solution
Service Catalog Item	Provides a standardized request form where employees enter software details such as Software Name, Version, Business Justification, Installation Device, License Type, and Required Installation Date.
Request (sc_request)	Acts as the parent record that groups all requested items submitted by an employee.
Requested Item (sc_req_item)	Tracks the complete lifecycle of each software installation request, including approvals, fulfillment, and closure.
Flow Designer	Automates manager approval routing, decision branching, task creation, status updates, and email notifications.
Approval Engine	Creates and manages approval records through the My Approvals interface.
Catalog Task (sc_task)	Automatically created after approval and assigned to the Network/Software Support Team for software installation.
Business Rule	Automatically creates an Incident whenever a software request cannot be fulfilled due to license unavailability.
Email Notifications	Sends automated notifications to employees regarding request submission, approval, rejection, and completion.

4. End-to-End Workflow

The proposed workflow consists of the following steps:

Step 1 – Request Submission

The employee accesses the **Service Catalog**, completes the Software Installation Request form, and submits the request.

Step 2 – Manager Approval

ServiceNow automatically routes the request to the employee's reporting manager. The manager reviews the request through the **My Approvals** workspace and either approves or rejects it.

Step 3 – Approved Request

If the request is approved:

- The request status is updated.
- A **Catalog Task** is automatically generated.

- The task is assigned to the **Network/Software Support Team**.
- The employee receives an email notification confirming the approval.

Step 4 – Rejected Request

If the request is rejected:

- The request status is updated to **Rejected**.
- The employee is notified by email along with the manager's comments.
- The workflow ends.

Step 5 – License Shortage Handling

If the requested software cannot be installed because the required license is unavailable:

- The request is placed in the **On Hold** state.
- A **Business Rule** automatically creates an **Incident**.
- The incident is assigned to the Software Support Team for investigation and resolution.

5. Benefits of the Proposed Solution

The proposed solution provides several operational and business benefits:

- Standardizes the software installation request process across the organization.
- Eliminates dependency on emails and manual request tracking.
- Reduces processing time through automated approval routing and task creation.
- Provides complete visibility into request status for employees, managers, and IT staff.
- Improves accountability by maintaining a complete audit trail of approvals and status changes.
- Ensures software installation requests are fulfilled in a structured and timely manner.
- Automatically identifies and escalates license shortages, reducing delays and manual follow-up.
- Supports future enhancements such as software license management, automated deployment, and multi-level approvals.

6. Alignment with Project Requirements

The proposed solution fully aligns with the functional and non-functional requirements defined for the project.

Functional Requirements

The solution satisfies the core functional requirements by:

- Providing a Service Catalog item for software installation requests.
- Capturing all required request information.
- Automatically routing requests for manager approval.
- Generating fulfillment tasks after approval.
- Sending automated email notifications.
- Creating incidents for software license shortages.
- Allowing employees to track request status throughout the workflow.

Non-Functional Requirements

The solution also meets important quality requirements by ensuring:

- **Usability:** A simple and user-friendly request form for employees.
- **Performance:** Fast workflow execution using Flow Designer.
- **Reliability:** Automated request processing without manual intervention.
- **Security:** Role-based access using ServiceNow Access Control Lists (ACLs).
- **Auditability:** Complete tracking of requests, approvals, and task activities.
- **Maintainability:** Modular implementation using native ServiceNow features such as Flow Designer and Business Rules.

Summary

The proposed **Employee Software Installation Request System** leverages ServiceNow's native capabilities to automate the complete request lifecycle—from request submission and manager approval to IT fulfillment and incident management. By replacing manual processes with an integrated workflow, the solution improves efficiency, transparency, and accountability while providing a scalable foundation for future enhancements such as license management, automated software deployment, and advanced approval workflows.

